Civil Engineering

Apply Algebra and Geometry

Learner Notes

Indices, logarithms, equations, coordinate geometry, areas, and engineering geometry.

Learning outcome	Focus
Apply laws of indices and logarithms	Applied mathematics in workplace and training contexts.
Solve algebraic and simultaneous equations	Applied mathematics in workplace and training contexts.
Apply coordinate geometry in civil layouts	Applied mathematics in workplace and training contexts.

1. Topic Overview

This topic develops mathematical skills required in Civil Engineering. Learners should show clear method, correct units, and technical interpretation.

2. Key Formulas

$$a^m \times a^n = a^{m+n}$$

$$\log(ab) = \log a + \log b$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

3. Worked-Example Method

Step 1: identify the given data. Step 2: select the correct formula. Step 3: substitute values carefully. Step 4: compute. Step 5: interpret the answer with correct units.

$$a^m \times a^n = a^{m+n}$$

4. Practice Questions

1. Solve equations arising from road gradient calculations.

2. Calculate distance between two control points.
3. Find the slope of a line joining two points.